Project: Worked examples
Title: Beam formulas (single span)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates reactions, maximum moment and shear, and maximum deflection of a single-span beam for the standard support and load cases. Left end at x = 0; loads downward; sagging moments and downward deflections positive.

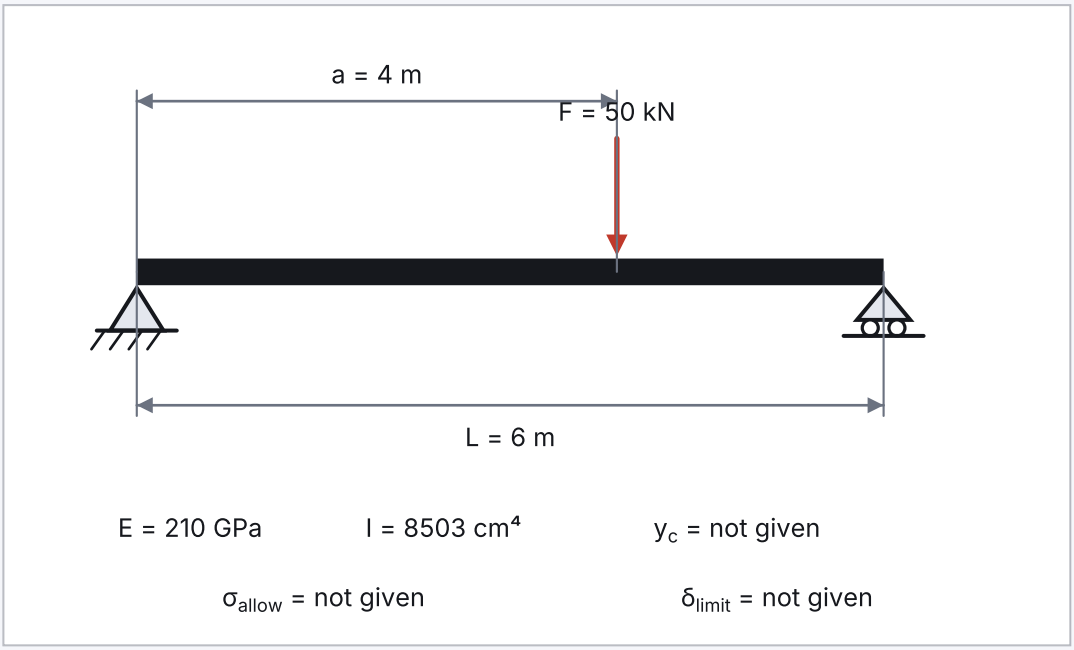


Figure 1 — Simply supported, point load F at a from the left end. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Formula source	Roark Table 8.1, in this sheet's own notation, for the chosen support and load case
Simply supported beam, point load at a (b = L − a), reaction and largest moment	$R_A = F\,b / L$; $M_{max} = F\,a\,b / L$
Simply supported beam, point load at a, largest deflection when a ≥ b	$\delta_{max} = F\,b\,(L^2 - b^2)^{(3/2)} / (9\sqrt{3}\,L\,EI)$ at $x = \sqrt{((L^2 - b^2)/3)}$
Propped cantilever, uniform load, reaction and moments	$R_B = 3wL/8$; $M_A = -wL^2/8$; sagging $M_{max} = 9wL^2/128$ at $x = 5L/8$

Basis: Shear, moment, slope and deflection formulas for elastic straight beams[1](Table 8.1, 8th ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
case	Case	simply supported, point load at a		assumption: Single span, pin–roller.	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
L	Span	6	m	drawing: DRG-FL-201 rev A	0.1 … 100 m
I	Second moment of area	8503	cm^4	reference: UB 305×165×40 (Blue Book)	1 … 100000000 cm^4
E	Young's modulus	210	GPa	spec: BS EN 1993-1-1 3.2.6	1 … 500 GPa
F	Point load, downward (optional; blank = not used by the distributed-load and couple cases)	50	kN	spec: Load schedule LS-01: plant load	0 … 100000 kN
a	Load position from the left end (optional; blank = not used by the distributed-load cases)	4	m	drawing: DRG-FL-201 rev A	0 … 100 m

3. Assumptions

A1. The beam is prismatic, linear elastic and Euler–Bernoulli, with small deflections. (Roark Table 8.1 conditions.)
A2. A cantilever is fixed at the left end (x = 0) and free at the right. (Case convention.)
A3. case (Case) is taken as simply supported, point load at a. (Single span, pin–roller.)

4. Working

[5] $EI = E * I$
$= 210\ GPa \times 8503\ cm^4$
$= 17856\ kN \cdot m^2$
Flexural rigidity

[8] $b = L - a$
$= 6\ m - 4\ m$
$= 2\ m$
Distance from the load to the right end

[9] $R_A = F * b / L$ [1](8.1 2a)
$= 50\ kN \times 2\ m / 6\ m$
$= 16.667\ kN$
Left reaction

[10] $R_B = F * a / L$
$= 50\ kN \times 4\ m / 6\ m$
$= 33.333\ kN$
Right reaction

[11] $M_{max} = F * a * b / L$
$= 50\ kN \times 4\ m \times 2\ m / 6\ m$
$= 66.667\ kN \cdot m$
Maximum moment, under the load

[12] $x_M = a$
$= 4\ m$
$= 4\ m$
Position of the maximum moment

[13] $M_{min} = 0\ kN*m$
$= 0\ kN \cdot m$
Minimum moment

[14] V_{max} = max(R_A, R_B)
= max(16.667 kN, 33.333 kN)
= 33.333 kN
Maximum shear
[15] c_{far} = b
= 2 m
= 2 m
Shorter segment (the deflection formula's b)
[16] x_{delta} = sqrt((L^2 - c_{far}^2) / 3)
= sqrt(((6 m)^2 - (2 m)^2) / 3)
= 3.266 m
Position of the maximum deflection
[17] δ_{max} = F * c_{far} * sqrt((L^2 - c_{far}^2)^3) / (9 * sqrt(3) * L * EI)
= 50 kN × 2 m × sqrt(((6 m)^2 - (2 m)^2)^3) / (9 × sqrt(3) × 6 m × 17856 kN·m²)
= 10.839 mm
Maximum deflection

5. Results

SYMBOL	VALUE	DESCRIPTION
R_A	16.67 kN	Left reaction
R_B	33.33 kN	Right reaction
M_{max}	66.67 kN·m	Maximum sagging moment
M_{min}	0 kN·m	Minimum (hogging) moment
V_{max}	33.33 kN	Maximum shear
δ_{max}	10.84 mm	Maximum deflection
x_M	4 m	Position of M_{max}
x_{delta}	3.266 m	Position of δ_{max}

6. Checks (governing first)

None.

7. Applicability

None.

8. Verification

Template struct.beam-formulas@1.2.0 Content hash f598af29a4da6e15 Engine calc-engine/1.0.0
Known-answer cases passed 8/8 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.